Lab Assignment 2: Python Data Visualization with Matplotlib, Pandas, and Seaborn

Objective: Gain hands-on experience using core Python visual libraries for univariate and bivariate analysis.
Dataset: Seaborn's built-in `tips` dataset (244 rows, 7 columns).
Tasks: (1) Matplotlib scatter plot of total_bill vs tip with custom colors, markers, labels and legend; (2) Pandas groupby + horizontal bar chart using `.plot(kind='barh')`; (3) Seaborn boxplot of total_bill by day with hue = sex.
Run with: `python lab2_python_data_visualization.py`

1. Python Code

```
# ======================================================================
# Lab Assignment 2: Python Data Visualization with Matplotlib, Pandas,
#                   and Seaborn
# ----------------------------------------------------------------------
# Objective : Hands-on experience using core Python visual libraries
#             for univariate and bivariate analysis.
# Dataset   : Seaborn's built-in "tips" dataset
# Run with  : python lab2_python_data_visualization.py
# Requires  : pip install matplotlib pandas seaborn
# Output    : Step-by-step console log + 3 PNG plots in ./plots/
# ======================================================================

import os
from datetime import datetime

import matplotlib
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns


# ---------- Helper: print a clearly visible step header --------------
def step(n, title):
    print("\n" + "=" * 70)
    print(f"STEP {n} : {title}")
    print("=" * 70)


# Save each figure as PNG; show it on screen too when a display exists
OUT_DIR = "plots"
os.makedirs(OUT_DIR, exist_ok=True)


def save_plot(fig, file):
    path = os.path.join(OUT_DIR, file)
    fig.savefig(path, dpi=110, bbox_inches="tight")
    print(">> Plot saved to:", path)
    if matplotlib.get_backend().lower() != "agg":
        plt.show()
    plt.close(fig)


pd.set_option("display.width", 100)
print("Lab Assignment 2 - Python Data Visualization (Matplotlib, Pandas, Seaborn)")
print("Python libraries : matplotlib", matplotlib.__version__,
      "| pandas", pd.__version__, "| seaborn", sns.__version__)
print("Run on           :", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))


# ======================================================================
# STEP 1 : Load the dataset
# ======================================================================
step(1, "Load the 'tips' dataset from Seaborn")

tips = sns.load_dataset("tips")
print(">> Dataset loaded successfully.")
print(">> Shape (rows x columns):", tips.shape)

```

```
print("\n>> Column data types:")
print(tips.dtypes)

print("\n>> First 5 rows:")
print(tips.head())

print("\n>> Missing values per column:")
print(tips.isnull().sum())

print("\n>> Summary of continuous variables:")
print(tips[["total_bill", "tip", "size"]].describe().round(2))


# ======================================================================
# STEP 2 : Matplotlib - Scatter plot of total_bill vs tip
# ======================================================================
step(2, "Matplotlib Scatter Plot: total_bill vs tip (colored by sex)")

r = tips["total_bill"].corr(tips["tip"])
print(f">> Pearson correlation between total_bill and tip: r = {r:.4f}")

# Customization: a different color and marker style for each group
styles = {"Male":   {"color": "royalblue", "marker": "o"},
          "Female": {"color": "crimson",   "marker": "^"}}
print(">> Plot customization used:")
for sex, st in styles.items():
    n = (tips["sex"] == sex).sum()
    print(f"   {sex:<6} -> color = {st['color']:<9}, marker = '{st['marker']}', points = {n}")

fig, ax = plt.subplots(figsize=(8, 5.5))
for sex, st in styles.items():
    subset = tips[tips["sex"] == sex]
    ax.scatter(subset["total_bill"], subset["tip"],
               c=st["color"], marker=st["marker"], s=45,
               alpha=0.7, edgecolors="white", linewidths=0.5,
               label=f"{sex} (n={len(subset)})")
ax.set_title("Total Bill vs Tip Amount (Tips dataset)", fontsize=13, fontweight="bold")
ax.set_xlabel("Total Bill (USD)", fontsize=11)
ax.set_ylabel("Tip (USD)", fontsize=11)
ax.grid(True, linestyle="--", alpha=0.4)
ax.legend(title="Customer Sex", loc="upper left")
save_plot(fig, "1_matplotlib_scatter_bill_vs_tip.png")

print(">> Interpretation: positive relationship - larger bills generally receive",
      "larger tips.")


# ======================================================================
# STEP 3 : Pandas - Group by a categorical variable + horizontal bars
# ======================================================================
step(3, "Pandas groupby('day') + Horizontal Bar Chart (.plot(kind='barh'))")

grouped = tips.groupby("day", observed=True).agg(
    num_bills=("total_bill", "count"),
    avg_total_bill=("total_bill", "mean"),
    avg_tip=("tip", "mean"),
)
print(">> Code: tips.groupby('day').agg(count, mean total_bill, mean tip)")
print(">> Grouped result:")
print(grouped.round(2))

avg_tip_by_day = grouped["avg_tip"]
print("\n>> Series plotted (average tip per day):")
print(avg_tip_by_day.round(2))

ax = avg_tip_by_day.plot(kind="barh",
                         color=["#4c72b0", "#55a868", "#c44e52", "#8172b2"],
                         edgecolor="black", figsize=(8, 5))
ax.set_title("Average Tip by Day of the Week", fontsize=13, fontweight="bold")
ax.set_xlabel("Average Tip (USD)", fontsize=11)
ax.set_ylabel("Day", fontsize=11)
for i, v in enumerate(avg_tip_by_day):
    ax.text(v + 0.03, i, f"${v:.2f}", va="center")
ax.set_xlim(0, avg_tip_by_day.max() * 1.15)
save_plot(ax.get_figure(), "2_pandas_barh_avg_tip_by_day.png")

```

```
print(f">> Interpretation: highest average tip on {avg_tip_by_day.idxmax()} "
      f"(${avg_tip_by_day.max():.2f}), lowest on {avg_tip_by_day.idxmin()} "
      f"(${avg_tip_by_day.min():.2f}).")


# ======================================================================
# STEP 4 : Seaborn - Boxplot of total_bill by day with hue = sex
# ======================================================================
step(4, "Seaborn Boxplot: total_bill by day, hue = sex (3rd dimension)")

print(">> Median total_bill for each day and sex:")
print(tips.pivot_table(values="total_bill", index="day", columns="sex",
                       aggfunc="median", observed=True).round(2))

# Outliers using the same 1.5 * IQR rule the boxplot whiskers use
print("\n>> Outliers detected (beyond 1.5 * IQR) in each day/sex group:")
found = False
for (day, sex), g in tips.groupby(["day", "sex"], observed=True):
    q1, q3 = g["total_bill"].quantile([0.25, 0.75])
    iqr = q3 - q1
    out = g[(g["total_bill"] < q1 - 1.5 * iqr) | (g["total_bill"] > q3 + 1.5 * iqr)]
    if len(out):
        found = True
        print(f"   {day:<4} {sex:<6}: {sorted(out['total_bill'].round(2).tolist())}")
if not found:
    print("   No outliers found.")

fig, ax = plt.subplots(figsize=(9, 5.5))
sns.boxplot(data=tips, x="day", y="total_bill", hue="sex",
            palette={"Male": "#6fa8dc", "Female": "#ea9999"},
            flierprops={"marker": "o", "markerfacecolor": "red", "markersize": 5},
            ax=ax)
ax.set_title("Total Bill by Day, split by Sex", fontsize=13, fontweight="bold")
ax.set_xlabel("Day of the Week", fontsize=11)
ax.set_ylabel("Total Bill (USD)", fontsize=11)
ax.legend(title="Sex")
save_plot(fig, "3_seaborn_boxplot_bill_by_day_sex.png")

print(">> Interpretation: weekend (Sat/Sun) bills are higher and more spread out;\n"
      "   male customers have higher median bills on 3 of the 4 days.")


# ======================================================================
# Done
# ======================================================================
step(5, "Lab Assignment 2 completed")
print("Files generated:")
for f in sorted(os.listdir(OUT_DIR)):
    print("  -", os.path.join(OUT_DIR, f))
```

2. Execution Output (console printout)

```
Lab Assignment 2 - Python Data Visualization (Matplotlib, Pandas, Seaborn)
Python libraries : matplotlib 3.11.2 | pandas 3.0.5 | seaborn 0.13.2
Run on           : 2026-10-26 21:28:08

======================================================================
STEP 1 : Load the 'tips' dataset from Seaborn
======================================================================
>> Dataset loaded successfully.
>> Shape (rows x columns): (244, 7)

>> Column data types:
total_bill     float64
tip            float64
sex           category
smoker        category
day           category
time          category
size             int64
dtype: object

>> First 5 rows:
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4

>> Missing values per column:
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64

>> Summary of continuous variables:
       total_bill     tip    size
count      244.00  244.00  244.00
mean        19.79    3.00    2.57
std          8.90    1.38    0.95
min          3.07    1.00    1.00
25%         13.35    2.00    2.00
50%         17.80    2.90    2.00
75%         24.13    3.56    3.00
max         50.81   10.00    6.00
======================================================================
STEP 2 : Matplotlib Scatter Plot: total_bill vs tip (colored by sex)
======================================================================
>> Pearson correlation between total_bill and tip: r = 0.6757
>> Plot customization used:
   Male   -> color = royalblue, marker = 'o', points = 157
   Female -> color = crimson  , marker = '^', points = 87
>> Plot saved to: plots/1_matplotlib_scatter_bill_vs_tip.png
>> Interpretation: positive relationship - larger bills generally receive larger tips.

======================================================================
STEP 3 : Pandas groupby('day') + Horizontal Bar Chart (.plot(kind='barh'))
======================================================================
>> Code: tips.groupby('day').agg(count, mean total_bill, mean tip)
>> Grouped result:
      num_bills  avg_total_bill  avg_tip
day
Thur         62           17.68     2.77
Fri          19           17.15     2.73
Sat          87           20.44     2.99
Sun          76           21.41     3.26

>> Series plotted (average tip per day):
day
Thur    2.77
Fri     2.73
```

```
Sat     2.99
Sun     3.26
Name: avg_tip, dtype: float64
>> Plot saved to: plots/2_pandas_barh_avg_tip_by_day.png
>> Interpretation: highest average tip on Sun ($3.26), lowest on Fri ($2.73).
======================================================================
STEP 4 : Seaborn Boxplot: total_bill by day, hue = sex (3rd dimension)
======================================================================
>> Median total_bill for each day and sex:
sex    Male  Female
day
Thur  16.98   13.78
Fri   17.22   15.38
Sat   18.24   18.36
Sun   20.72   17.41

>> Outliers detected (beyond 1.5 * IQR) in each day/sex group:
   Thur Male  : [41.19]
   Thur Female: [29.8, 34.83, 43.11]
   Sat  Male  : [48.27, 48.33, 50.81]
   Sat  Female: [44.3]
   Sun  Male  : [45.35, 48.17]
>> Plot saved to: plots/3_seaborn_boxplot_bill_by_day_sex.png
>> Interpretation: weekend (Sat/Sun) bills are higher and more spread out;
   male customers have higher median bills on 3 of the 4 days.

======================================================================
STEP 5 : Lab Assignment 2 completed
======================================================================
Files generated:
  - plots/1_matplotlib_scatter_bill_vs_tip.png
  - plots/2_pandas_barh_avg_tip_by_day.png
  - plots/3_seaborn_boxplot_bill_by_day_sex.png
```

3. Output Plots

Figure 1: Matplotlib Scatter Plot - Total Bill vs Tip (color/marker by sex)

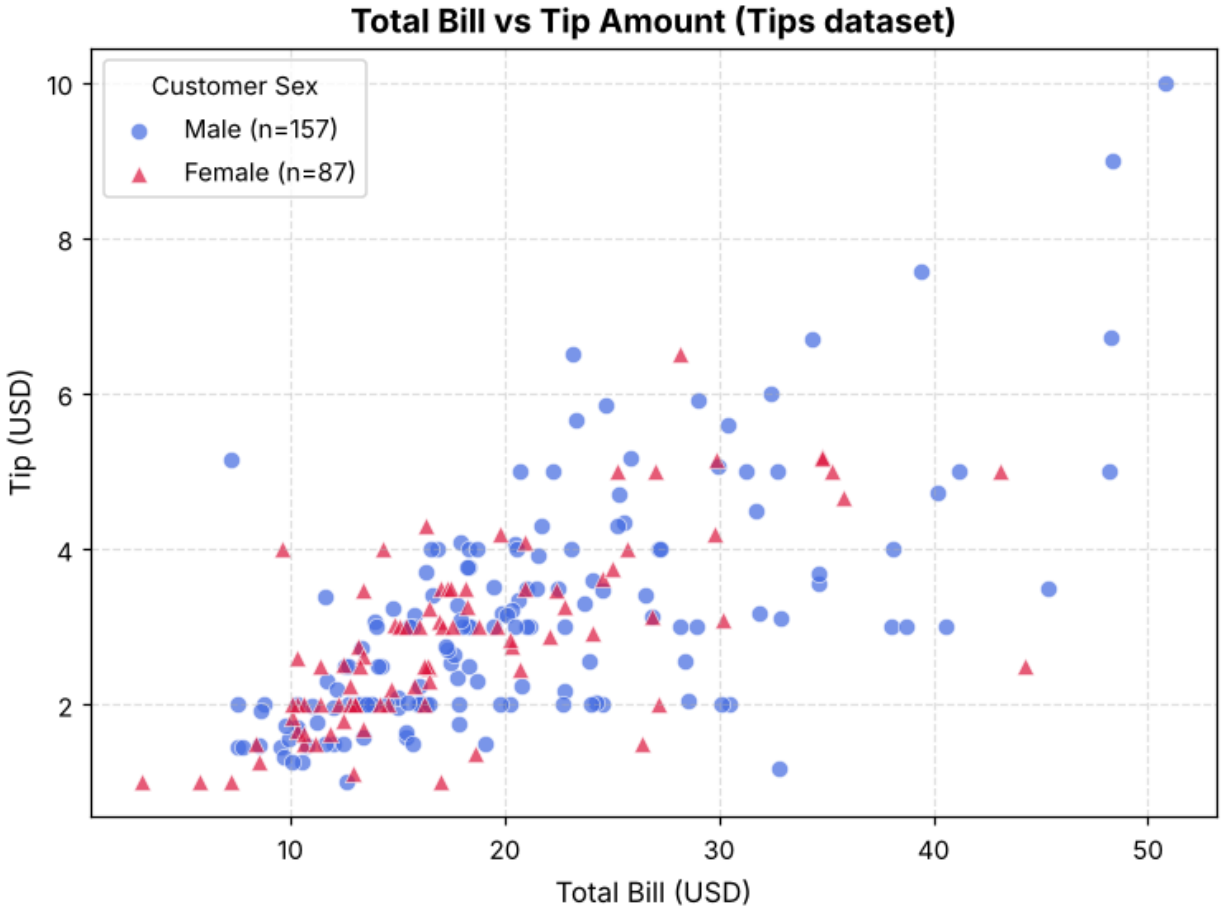


Figure 2: Pandas Horizontal Bar Chart - Average Tip by Day

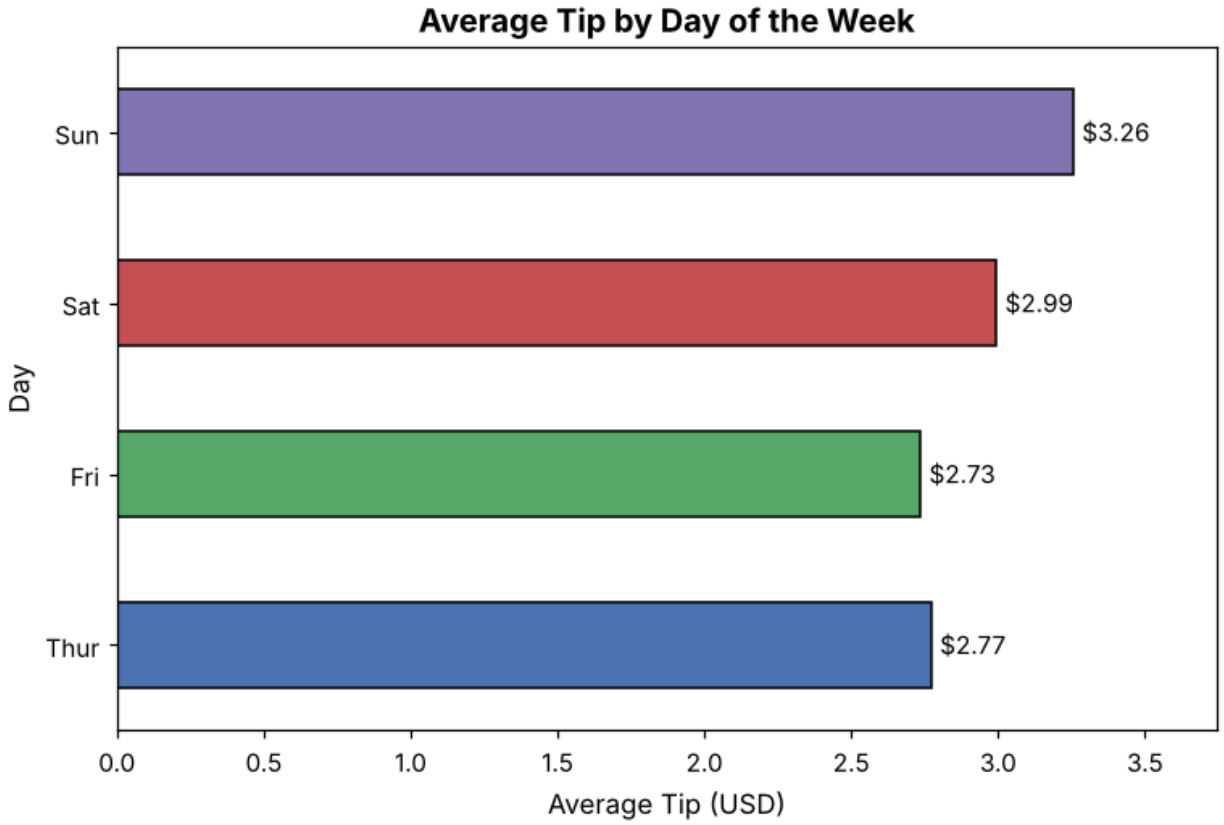


Figure 3: Seaborn Boxplot - Total Bill by Day, hue = Sex

